Blockchain Technology Practice Questions

1. Explain any three important properties of blockchain technology and their significance.

2. Differentiate between a centralized database and a blockchain with respect to data management, control, and security.

3. Explain the structure of a blockchain block with a neat diagram, showing the block header, transaction data, and previous block hash.

4. What is block mining? Explain the major steps involved in mining a block in a Proof-of-Work blockchain.

5. What is a distributed ledger? How does it differ from a conventional database?

6. Explain the process of transaction propagation and block propagation in a peer-to-peer blockchain network.

7. Discuss any four common myths about blockchain technology and explain the actual limitations or conditions associated with each.

8. Differentiate between private-key cryptography and public-key cryptography with suitable examples.

9. Explain the encryption and decryption process using public-key cryptography with a suitable example.

10. Define a cryptographic hash function and explain any four important properties required for blockchain security.

11. What is a message digest? Explain the role of SHA-256 in blockchain technology.

12. Explain the concept of a hash pointer. How does it help maintain the integrity of blockchain data?

13. Explain the construction of a Merkle tree for four transactions with a neat diagram. What is the significance of the Merkle root?

14. Describe the construction of a hash chain. Explain how a hash chain helps detect modifications to previously stored blocks.

15. In an RSA system, let p=5, q=11, and the public exponent e=3. Calculate n, phi(n), and the private exponent d. Show the necessary steps.

16. Explain the process of creating and verifying a digital signature using public-key cryptography.

17. Differentiate between encryption and digital signatures in terms of confidentiality, authentication, and integrity.

18. What are the different types of hashing? Explain the importance of cryptographic hashing in blockchain technology.

19. Explain how a Merkle tree allows a transaction to be verified without downloading every transaction in a block. Illustrate your answer with an example.

20. Define a distributed system. Explain how blockchain technology uses the principles of distributed systems.

21. What is distributed consensus? Why is it necessary for maintaining a consistent blockchain ledger?

22. Explain the evolution of cryptocurrencies from Bitcoin to blockchain platforms supporting decentralized applications.

23. Discuss the major design goals of cryptocurrencies.

24. Explain how Bitcoin prevents the double-spending problem without depending on a central financial authority.

25. What is a cryptocurrency wallet? Differentiate between desktop wallets, mobile application-based wallets, and browser-based wallets.

26. Explain the role of public and private keys in a cryptocurrency wallet. Why is protecting the private key essential?

27. Describe the steps involved in transferring cryptocurrency using MetaMask. Explain the importance of selecting the correct network and verifying the recipient's address.

28. Explain the Bitcoin mining process and discuss the role of miners in transaction validation and block creation.

29. What is a decentralized application (DApp)? Compare DApps with conventional applications in terms of architecture, control, and data management.

30. Differentiate between permissionless and permissioned blockchain models, giving suitable examples.

31. What is the consensus problem in distributed systems? Explain two major challenges in achieving consensus.

32. Explain the Byzantine Generals problem and its relevance to blockchain consensus mechanisms.

33. Explain Nakamoto consensus and the role of Proof of Work in selecting the accepted blockchain.

34. Discuss the major limitations of Proof of Work (PoW).

35. What is a 51% attack? Explain its potential consequences for a Proof-of-Work blockchain.

36. Compare Proof of Stake (PoS), Proof of Activity (PoAc), Proof of Burn (PoB), Proof of Authority (PoAu), Proof of Elapsed Time (PoET), and Proof of Importance (PoI).

37. Explain the role of Ethereum in blockchain technology. How does it support decentralized applications?

38. What is a smart contract? Explain how smart contracts are developed using Solidity and executed on Ethereum.

39. Explain the Two Generals problem and discuss why reliable agreement is difficult when messages may be lost.

40. What is the Byzantine agreement problem? How does it differ from the Two Generals problem?

41. Explain the working of Practical Byzantine Fault Tolerance (PBFT), describing its major phases.

42. Differentiate between Proof of Work and PBFT in terms of participant requirements, communication, and application scenarios.

43. Explain the architecture of Hyperledger Fabric. Discuss the roles of peers, ordering services, organizations, and chaincode in a permissioned blockchain.

44. Explain how blockchain can improve transparency and traceability in land registry records.

45. Discuss four potential governmental applications of blockchain technology.

46. Explain the working of a blockchain-based decentralized marketplace and identify its major advantages.

47. How can blockchain technology facilitate cross-border payments? Discuss its advantages and limitations.

48. Explain how a blockchain-based Know Your Customer (KYC) system can support secure identity verification among multiple financial institutions.

49. Discuss the potential use of blockchain in academic certificate verification. How can it help reduce fraudulent certificates?

50. A government wants to use blockchain for public-benefit distribution. Explain how blockchain can improve accountability and auditability, and discuss the challenges related to privacy, incorrect data, and implementation costs.